Lab_sheet-02

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Section-c

Roll no-11

Experiment No. 6

Experiment Name:

Contrast Enhancement and Histogram-Based Image Processing using Python and OpenCV

Step 1: Import Required Libraries

In [1]:
```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

Step 2: Load and Convert Image to Grayscale

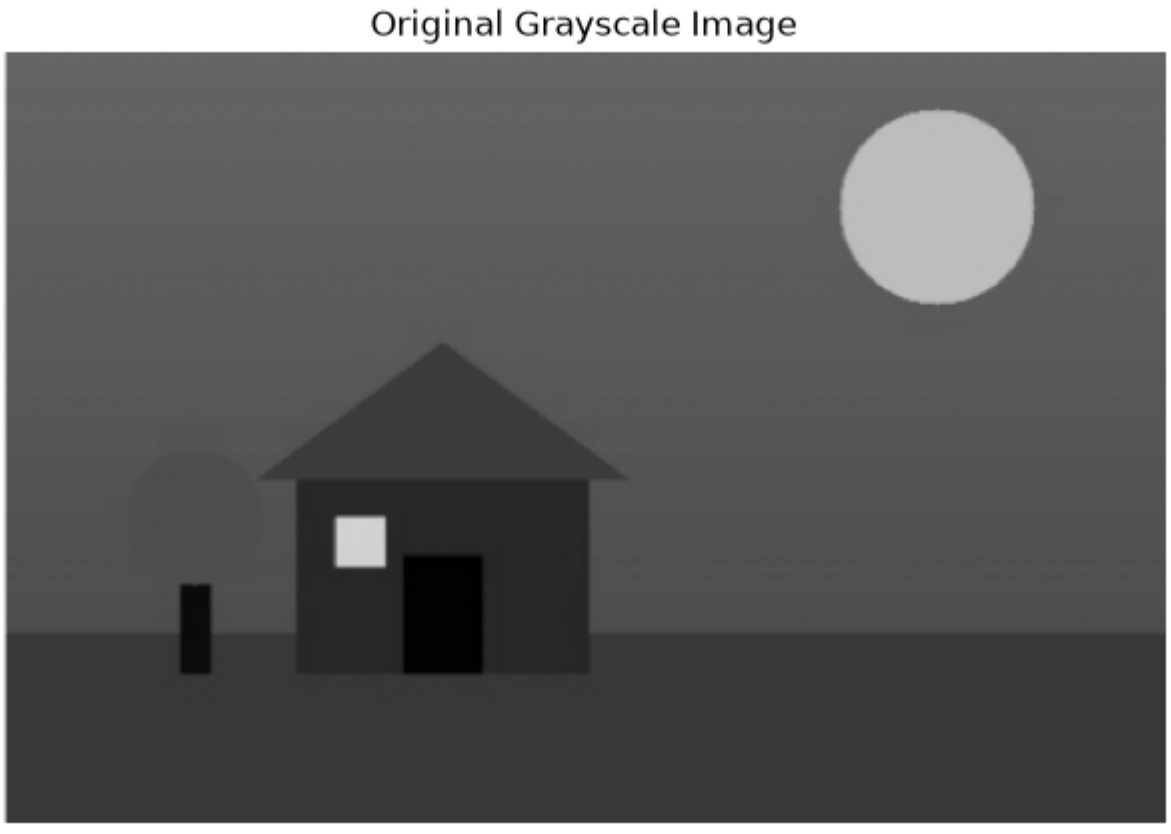
In [2]:
```
img = cv2.imread("my_pic01.png")

if img is None:
    print("Image not found!")
else:
    print("Image loaded successfully.")

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(8, 5))
plt.imshow(gray, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

```
Image loaded successfully.
```



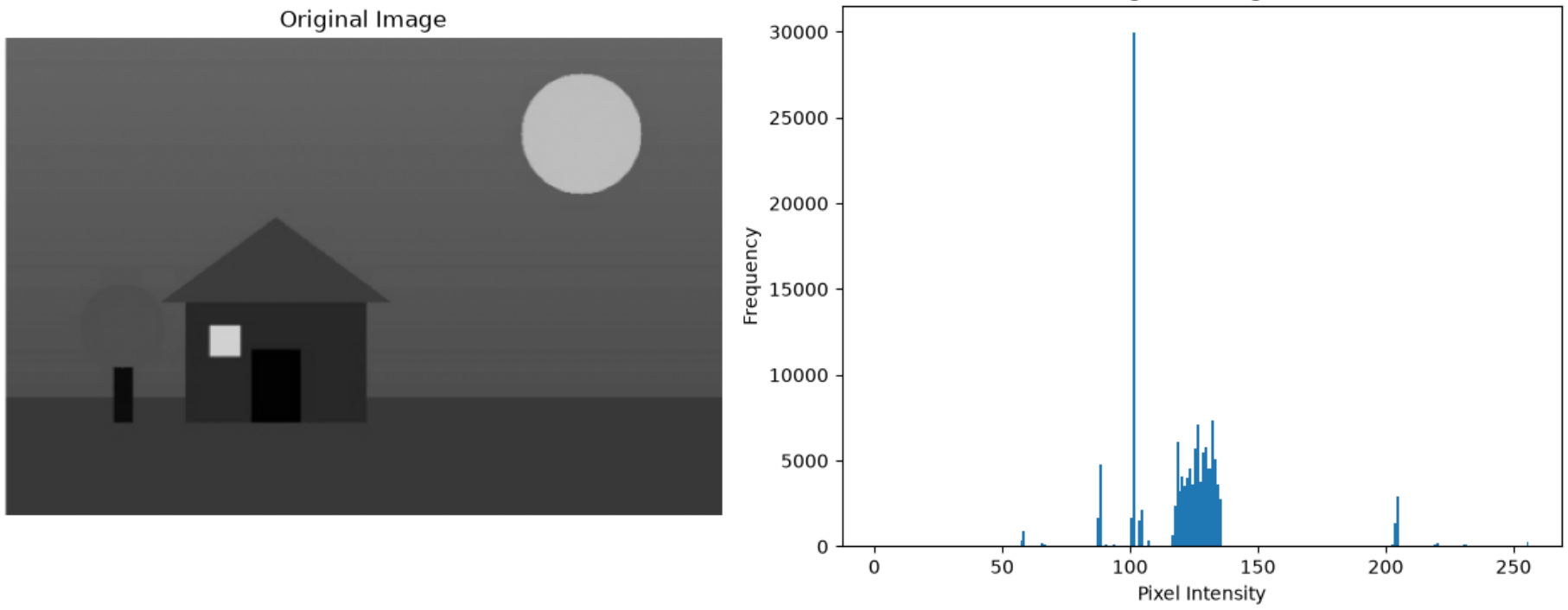
Step 3: Display Original Image and Histogram

In [3]:
```
plt.figure(figsize=(12, 5))

plt.subplot(1, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```

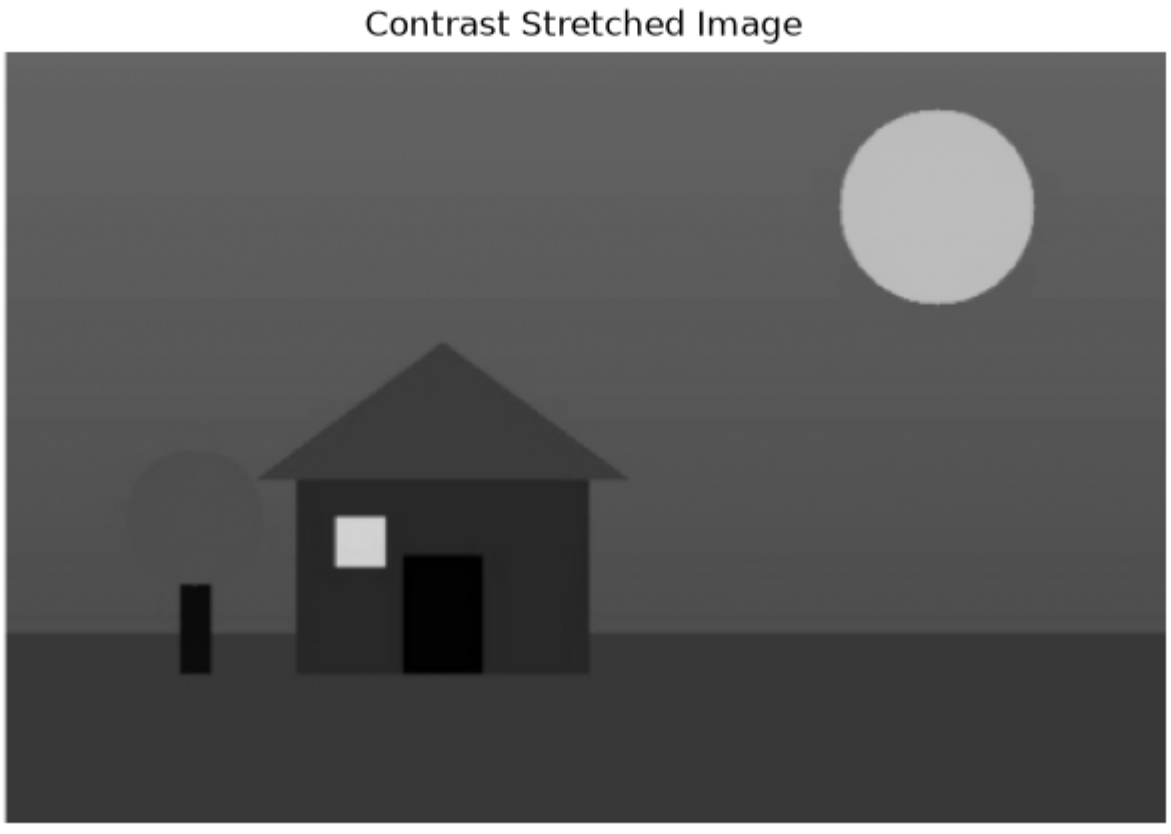


Step 4: Contrast Stretching

In [4]:
```
min_val = np.min(gray)
max_val = np.max(gray)

contrast_stretched = ((gray - min_val) / (max_val - min_val) * 255).astype(np.uint8)

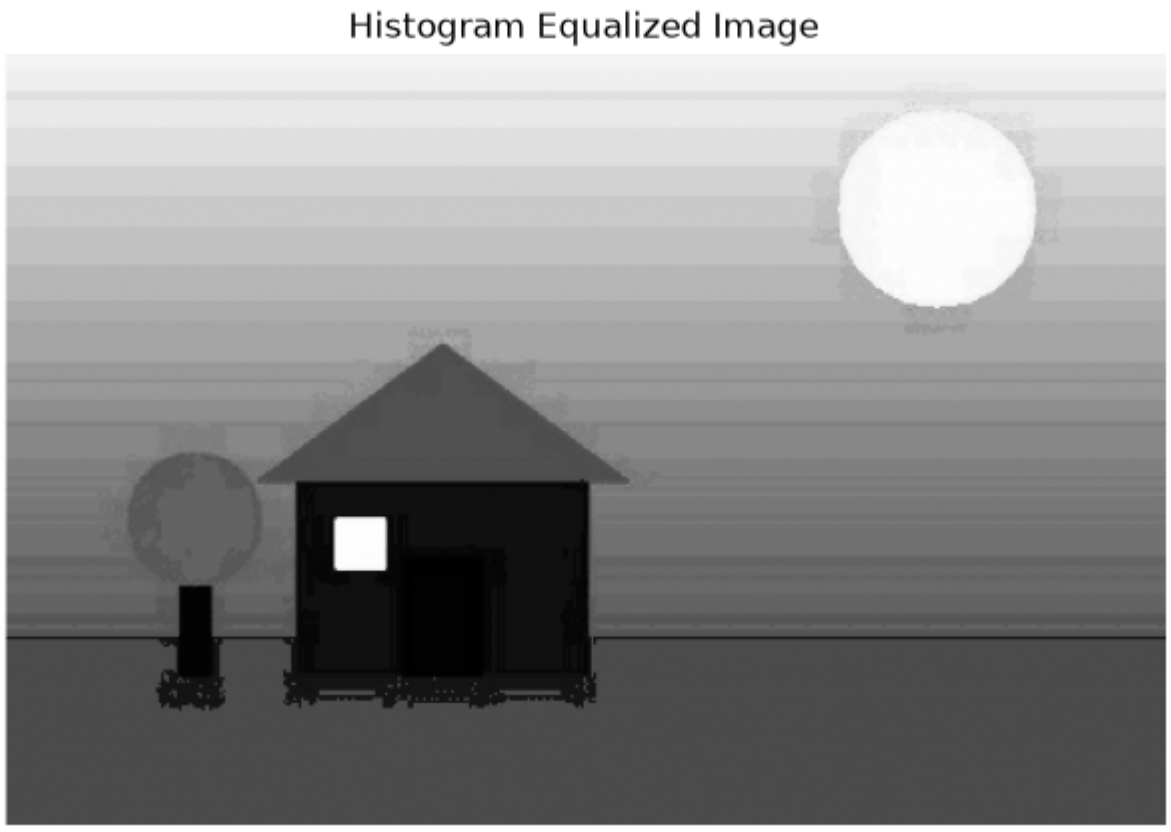
plt.figure(figsize=(8, 5))
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretched Image")
plt.axis("off")
plt.show()
```



Step 5: Histogram Equalization

In [5]:
```
hist_equalized = cv2.equalizeHist(gray)

plt.figure(figsize=(8, 5))
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalized Image")
plt.axis("off")
plt.show()
```

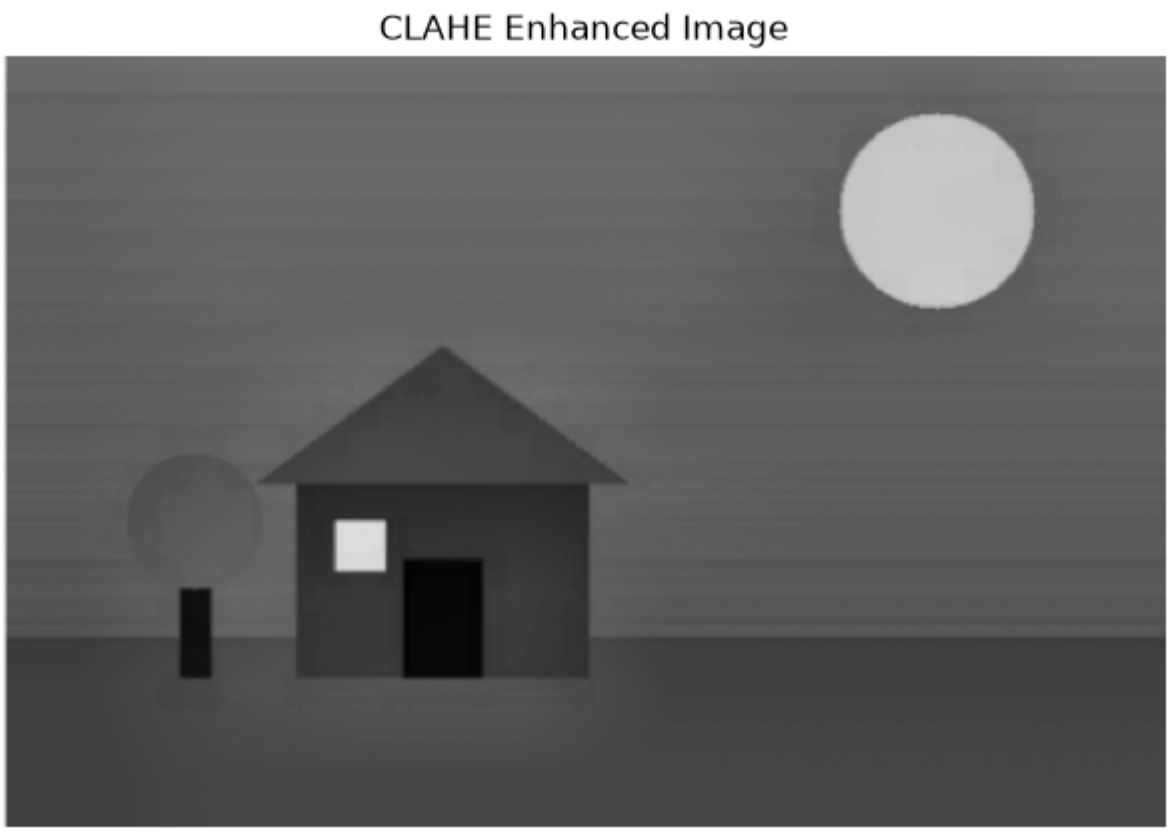


Step 6: Apply CLAHE

In [6]:
```
clahe = cv2.createCLAHE(
    clipLimit=2.0,
    tileGridSize=(8, 8)
)

clahe_image = clahe.apply(gray)

plt.figure(figsize=(8, 5))
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE Enhanced Image")
plt.axis("off")
plt.show()
```



Step 7: Compare Histograms

In [7]:
```
plt.figure(figsize=(15, 5))

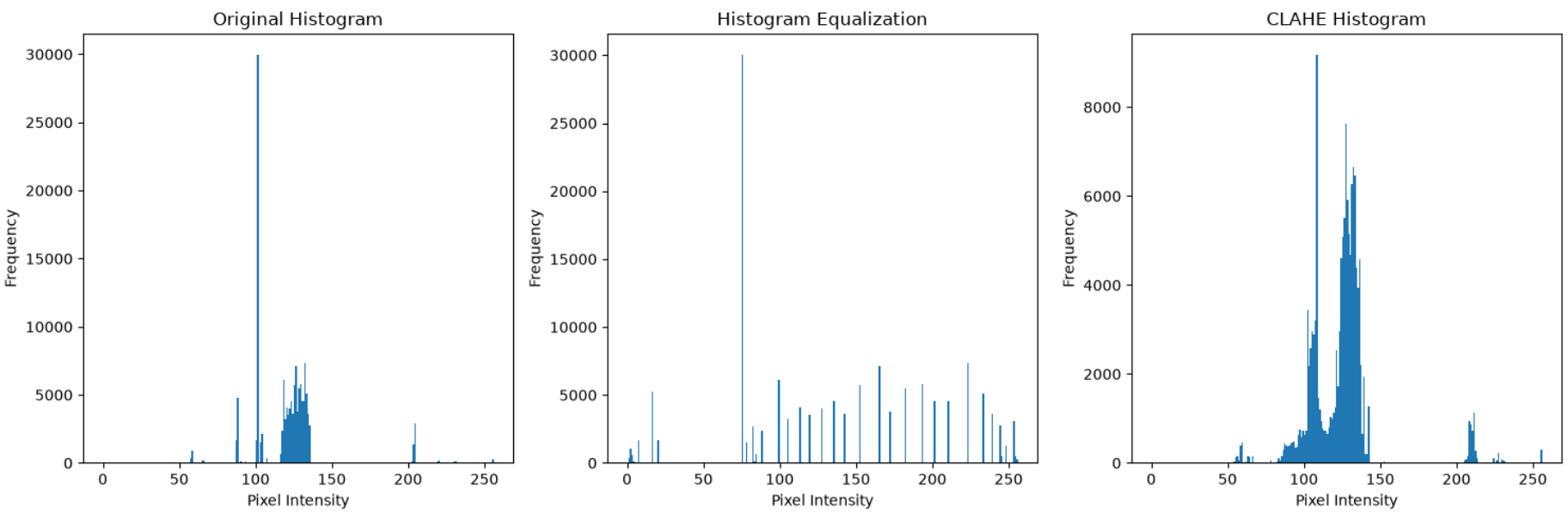
plt.subplot(1, 3, 1)
plt.hist(gray.ravel(), bins=256, range=[0, 256])
plt.title("Original Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 3, 2)
plt.hist(hist_equalized.ravel(), bins=256, range=[0, 256])
```

```
plt.title("Histogram Equalization")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.subplot(1, 3, 3)
plt.hist(clahe_image.ravel(), bins=256, range=[0, 256])
plt.title("CLAHE Histogram")
plt.xlabel("Pixel Intensity")
plt.ylabel("Frequency")

plt.tight_layout()
plt.show()
```



Step 8: Compare All Enhanced Images

In [8]:
```
plt.figure(figsize=(15, 10))

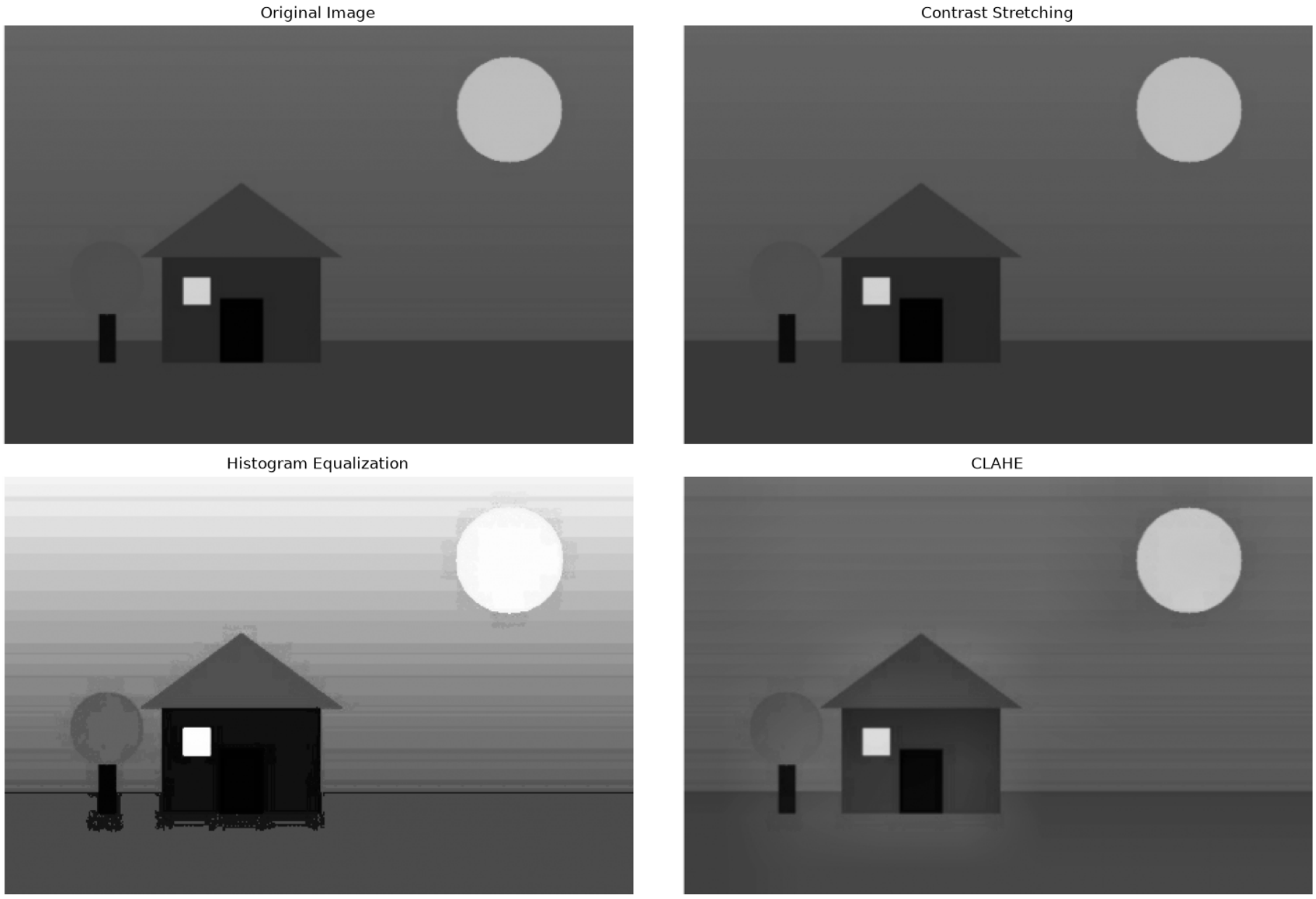
plt.subplot(2, 2, 1)
plt.imshow(gray, cmap="gray")
plt.title("Original Image")
plt.axis("off")

plt.subplot(2, 2, 2)
plt.imshow(contrast_stretched, cmap="gray")
plt.title("Contrast Stretching")
plt.axis("off")

plt.subplot(2, 2, 3)
plt.imshow(hist_equalized, cmap="gray")
plt.title("Histogram Equalization")
plt.axis("off")

plt.subplot(2, 2, 4)
plt.imshow(clahe_image, cmap="gray")
plt.title("CLAHE")
plt.axis("off")

plt.tight_layout()
plt.show()
```



In [9]:
```
print("IMAGE CONTRAST ENHANCEMENT ANALYSIS")
print("-----------------------------------")

print("1. Original Image:")
print("   May have low contrast and limited intensity distribution.")

print("\n2. Contrast Stretching:")
print("   Expands the intensity range and improves contrast.")

print("\n3. Histogram Equalization:")
print("   Redistributes pixel intensities to improve global contrast.")

print("\n4. CLAHE:")
print("   Performs local contrast enhancement while limiting noise amplification.")
```

```
IMAGE CONTRAST ENHANCEMENT ANALYSIS
-----------------------------------
1. Original Image:
   May have low contrast and limited intensity distribution.

2. Contrast Stretching:
   Expands the intensity range and improves contrast.

3. Histogram Equalization:
   Redistributes pixel intensities to improve global contrast.

4. CLAHE:
   Performs local contrast enhancement while limiting noise amplification.
```

In [10]:
```
print("CONCLUSION")
print("-----------")
print("Contrast enhancement techniques were successfully implemented.")
print("Contrast stretching improves the dynamic intensity range.")
print("Histogram Equalization improves global image contrast.")
print("CLAHE provides local contrast enhancement and is useful for")
print("images with varying illumination.")
```

```
CONCLUSION
-----------
Contrast enhancement techniques were successfully implemented.
Contrast stretching improves the dynamic intensity range.
Histogram Equalization improves global image contrast.
CLAHE provides local contrast enhancement and is useful for
images with varying illumination.
```

Questions

1. What is image contrast, and why is it important?

Answer: Image contrast is the difference between bright and dark regions. It improves image visibility and feature clarity.

2. What is an image histogram?

Answer: A histogram shows the frequency distribution of pixel intensity values in an image.

3. Histogram Stretching vs Histogram Equalization

Answer: Stretching expands the intensity range, while equalization redistributes intensities to improve contrast.

4. What is CLAHE?

Answer: CLAHE improves local contrast in small regions and limits noise amplification. Standard equalization works globally.

5. Why use histogram equalization before feature extraction and segmentation?

Answer: It improves contrast and makes important features more visible for further image processing.

6. Three applications of histogram equalization

Answer: Medical imaging, satellite imaging, and low-light image enhancement.

7. Limitations of global histogram equalization

Answer: It can amplify noise and may not perform well with uneven illumination.

8. How does contrast enhancement improve object detection?

Answer: It makes object boundaries and features clearer, helping detection and recognition systems.

9. Histogram enhancement vs brightness adjustment

Answer: Brightness adjustment changes overall intensity, while histogram enhancement changes intensity distribution to improve contrast.

10. Why is CLAHE preferred for medical and low-light images?

Answer: CLAHE improves local contrast and handles uneven illumination while limiting excessive enhancement.